

Supplemental Materials

An Empirical Study of Quality of Change-level Feature in JIT-SDP

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Abstract—This document presents additional information of the submitted paper “An Empirical Study of Quality of Change-level Feature in JIT-SDP”.

I. RQ1: IMPACT OF BRANCH DEPENDENCY ON FEATURE NOISE

In our paper, we have shown the noise rate of the 14 features due to the lack of branch dependency information. In addition to this, we also used other vector similarity evaluation indicators to calculate the difference between the two sets of features (F^- denotes the features extracted ignoring the branch dependency information and F^+ indicates the features considering the brach dependency information) which can be checked in Table I and II.

The results of Spearman correlation coefficient have shown that the datasets are indeed affected by the lack of branch dependency information, and some items are affected by it seriously. However, the results of Pearson Correlation have shown that the distribution of the datasets are not affected seriously.

TABLE I: Spearman Correlation

Features Datasets	LT	NDEV	AGE	NUC	REXP
Brackets	0.4965	0.4961	0.4533	0.4497	0.7788
Broadleaf	0.5596	0.5491	0.4862	0.5813	0.7863
Camel	0.9656	0.9556	0.9638	0.9740	0.9925
Corefx-v-6.2	0.6385	0.7947	0.6217	0.7006	0.7860
Django	0.9841	0.9900	0.9855	0.9828	0.9876
Elasticsearch	0.4958	0.1301	0.2709	0.4042	0.6845
Fabric8	0.9562	0.9941	0.9140	0.9588	0.9480
Godot	0.5512	0.4017	0.5175	0.4394	0.5709
Vscode	0.5715	0.5757	0.5552	0.5720	0.8172
Pytorch	0.9860	0.9994	0.9750	0.9904	0.9882
Spring-boot	0.4359	0.1889	0.3911	0.3893	0.7482
Node	0.9207	0.9925	0.9501	0.9171	0.9466
Nova	0.5587	0.4716	0.5959	0.5210	0.6982
Tomcat	0.9991	0.9965	0.9929	0.9979	0.9949
Pandas	0.8921	0.9927	0.9395	0.9158	0.9655
Security	0.8552	0.6650	0.6966	0.8141	0.8605
Rails	0.7506	0.6907	0.6859	0.7086	0.8801
Rust	0.3813	0.0250	0.2197	0.2714	0.7374
Wp-calypto	0.9227	0.9871	0.9404	0.9523	0.9491
Npm	0.9770	0.9823	0.9817	0.9782	0.9904
Mybatis3	0.9050	0.8691	0.8403	0.9105	0.9382
Tensorflow	0.5285	0.7796	0.5154	0.6156	0.7359

II. RQ2: IMPACT OF FEATURE NOISE ON PREDICTION PERFORMANCE

In our paper, we have conducted a linear significance analysis of these 14 features using Logistic Regression; As

TABLE II: Pearson Correlation

Features Datasets	LT	NDEV	AGE	NUC	REXP
Brackets	0.8159	0.6332	0.8705	0.7678	0.9425
Broadleaf	0.7199	0.6741	0.7566	0.8571	0.9729
Camel	0.9983	0.9670	0.9872	0.9881	0.9942
Corefx-v-6.2	0.7034	0.8090	0.9086	0.9886	0.6983
Django	0.9933	0.9993	0.9977	0.9979	0.9998
Elasticsearch	0.9184	0.3528	0.5642	0.9055	0.7382
Fabric8	0.9957	0.9905	0.9858	0.9998	0.9988
Godot	0.9033	0.5474	0.7956	0.9254	0.9970
Vscode	0.8493	0.6680	0.7848	0.8005	0.9558
Pytorch	0.9983	0.9997	0.9923	0.9998	0.9999
Spring-boot	0.5976	0.3813	0.6278	0.5862	0.9988
Node	0.9844	0.9954	0.9882	0.9754	0.9996
Nova	0.8167	0.6152	0.8277	0.8089	0.9897
Tomcat	0.9998	0.9958	0.9987	0.9990	0.9985
Pandas	0.9672	0.9984	0.9891	0.9891	0.9987
Security	0.9238	0.7430	0.8259	0.9822	0.9508
Rails	0.8938	0.7779	0.8651	0.8556	0.9971
Rust	0.5974	0.2375	0.4709	0.6445	0.9141
Wp-calypto	0.9903	0.9866	0.9900	0.9971	0.9930
Npm	0.9994	0.9989	0.9984	0.9975	0.9996
Mybatis3	0.9331	0.9007	0.9348	0.9828	0.3788
Tensorflow	0.8704	0.7825	0.8904	0.8458	0.9936

shown in Fig III, we checked the significance of the 14 features across 22 projects. We can find that the features REXP is not significant in many projects, but the other features are almost significant in the 22 projects. This means that the features we are targeting are effective, which makes our experiment more valid.

III. RQ3: IMPACT OF FEATURE NOISE ON MODEL RANKING

We have evaluated the rank score of the 5 models both for the two datasets (F^- and F^+) and used box plots to display and analyze the kendall tau [1] in our paper. Here we will display more detailed information about our results of the rank score. As shown in Fig 1, we compared the final Rank scores of the five models on F^- and F^+ both for G-Mean and MCC.

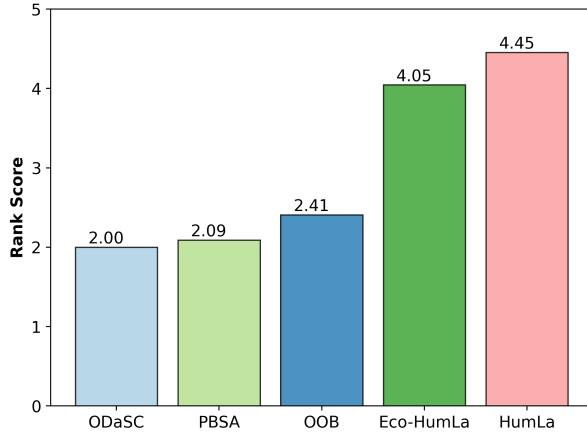
The results indicate that, in terms of specific numerical values, the rank score has indeed been affected, and we have observed some changes in the arrangement of certain models.

REFERENCES

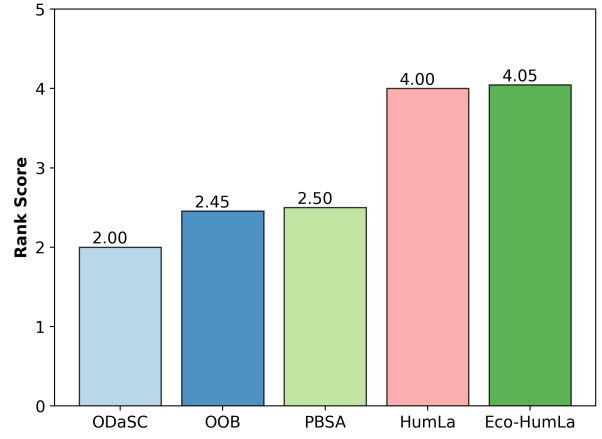
- [1] M. G. Kendall, “A new measure of rank correlation,” *Biometrika*, vol. 30, no. 1-2, pp. 81–93, 1938.

TABLE III: The statistical significance of 14 features across 22 projects. We set up that a p -value less than 0.05 indicates that the feature has statistical significance, with '✓' representing significant and '✗' representing not significant.

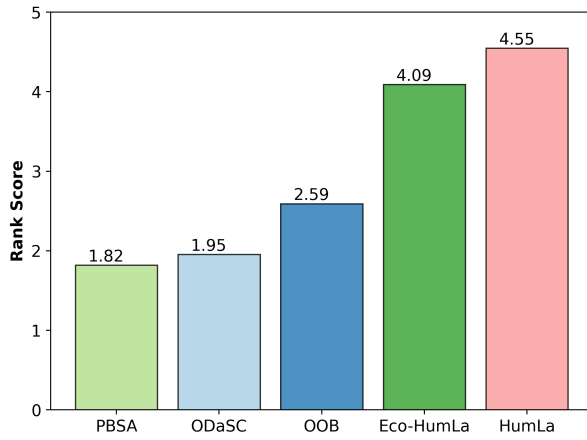
[illegible]



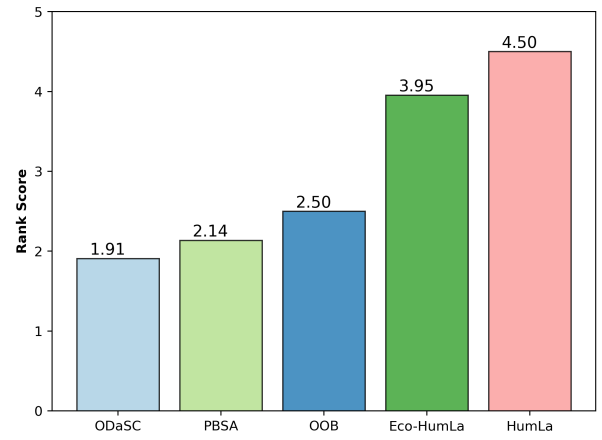
(a) G-Mean Rank Score on noisy feature values F^-



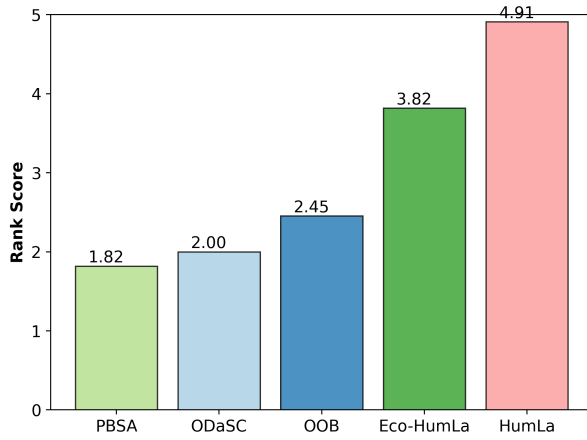
(b) G-Mean Rank Score on new feature values F



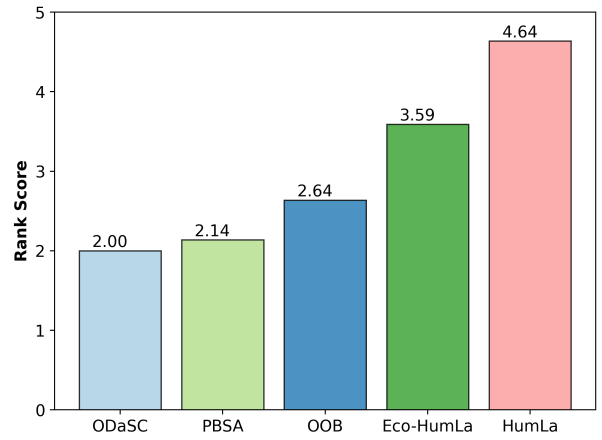
(c) MCC Rank Score on noisy feature values F^-



(d) MCC Rank Score on new feature values F



(e) F1-Score Rank Score on noisy feature values F^-



(f) F1-Score Score on new feature values F

Fig. 1: Rank Score for 5 Models